

Presentation Of The Final Year Project

Design and Implementation of an AI-Powered Study Hub

Represented By

Mouhamed Gharsallah & Tony Charmant Egerimana

Supervisor

Courinne Amel Zayani

Faculty of Sciences of Sfax, University of Sfax – June 2026

Problematic

1 Scattered & Unofficial

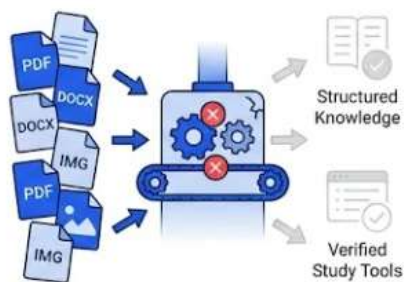
SCATTERED ACROSS UNOFFICIAL CHANNELS



Content is hard to find and easy to lose.

2 Broken Pipeline

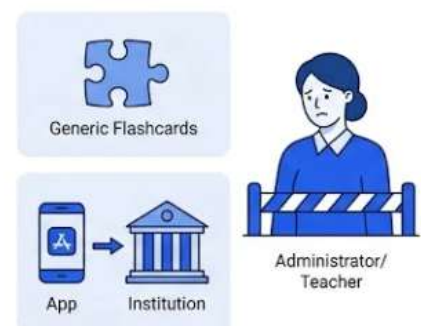
NO STANDARD TRANSFORMATION PIPELINE



Raw files do not automatically convert into structured, searchable tools.

3 Fragmented & Unmanaged

EXISTING TOOLS LACK INTEGRATION



Tools operate outside the framework; no institutional quality control.

Existing Solutions – Limitations

CAPABILITY	Google NotebookLM	Adobe Student Spaces
AI Chat and Generated Study Assets	✓	✓
Multi-level institutional management	✗	✗
Teacher roles for uploading course materials	✗	✗
Adaptive Intelligent Tutoring System (ITS)	✗	✗

➡ **Google NotebookLM** and **Adobe Student Spaces** are powerful individual study tools. They are not designed to be deployed as a governed, multi-role institutional platform.

Presentation Outline

- 2 Methodology
- 3 Architecture & Tech Stack
- 4 Core Features
- 5 Demonstration
- 6 Validation & Results
- 7 Future Work & Conclusion

Methodology

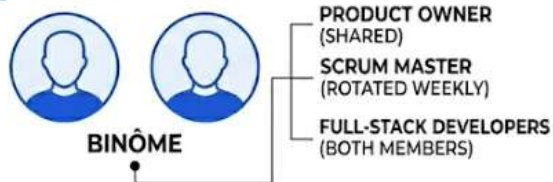
SPRINTS



USER STORIES & MOSCOW



ROLES

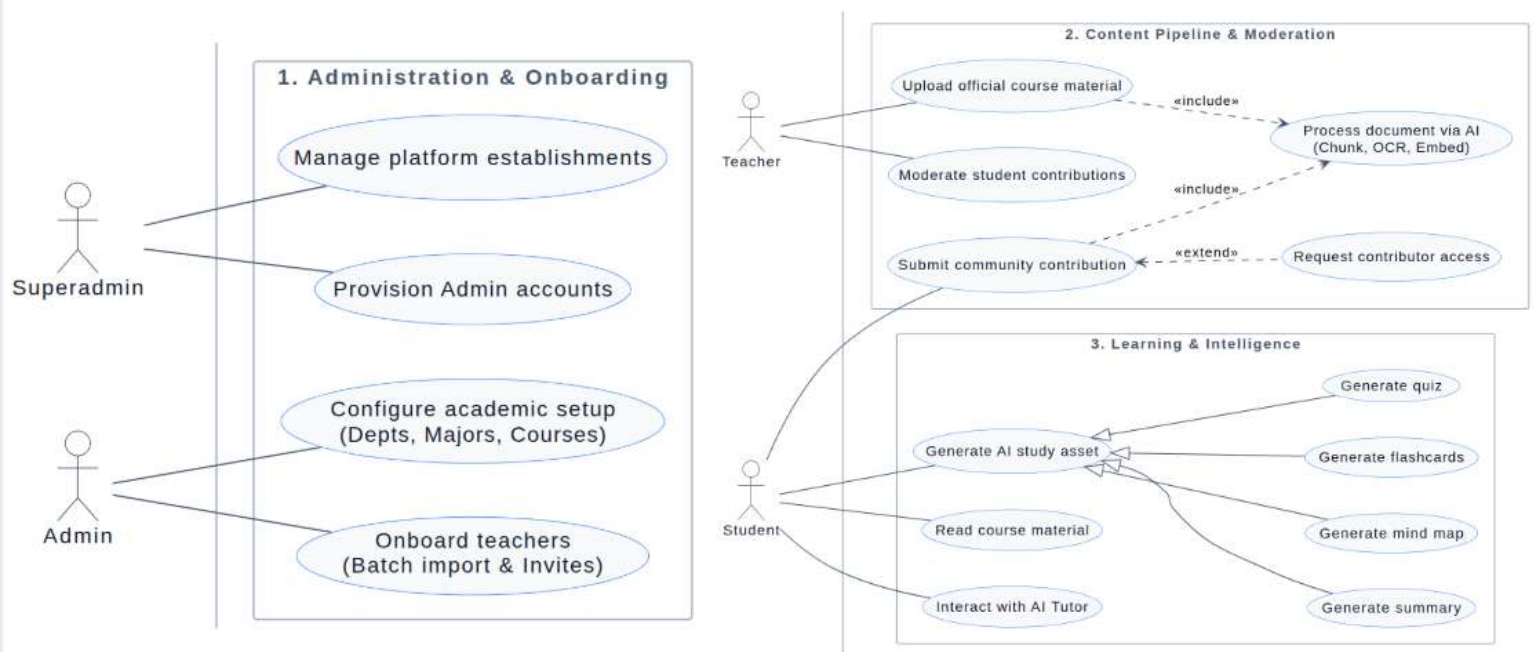


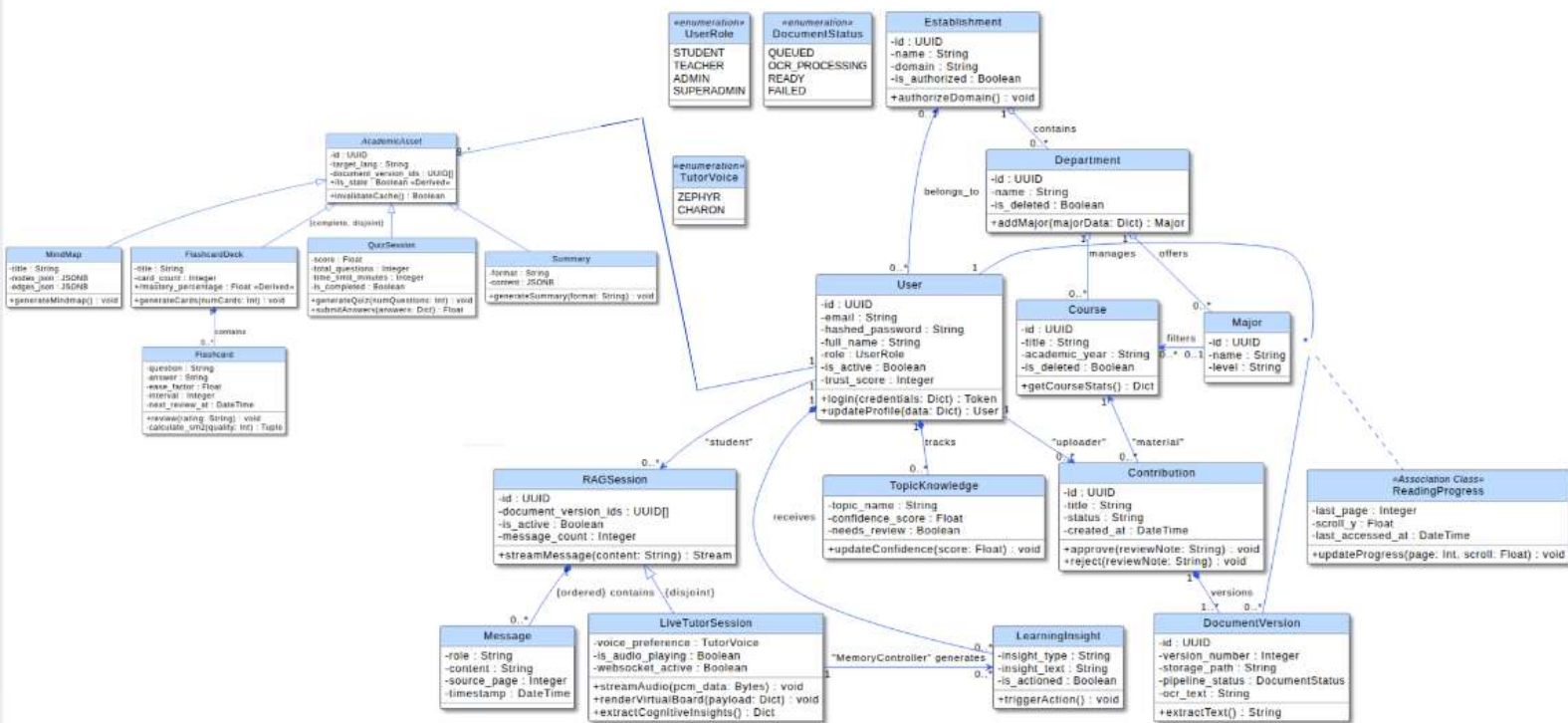
CEREMONIES

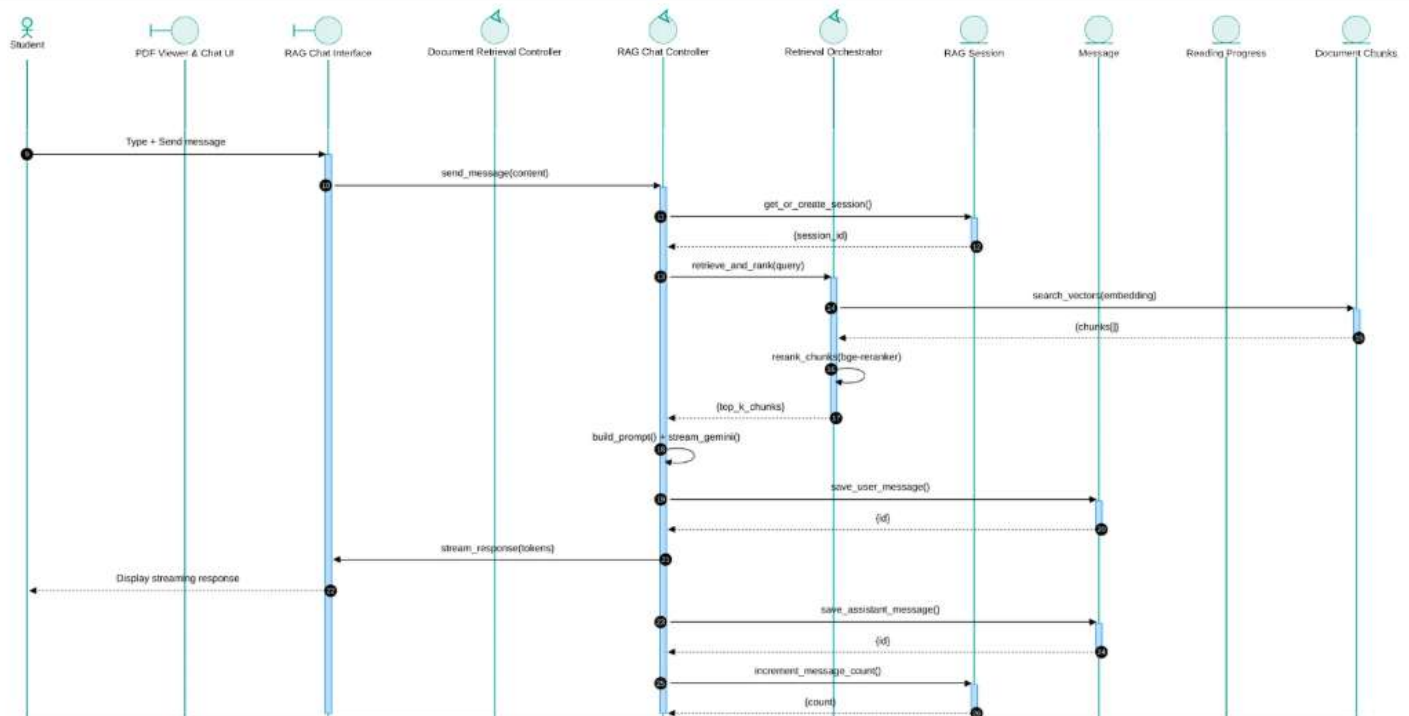


DEFINITION OF DONE (DoD)

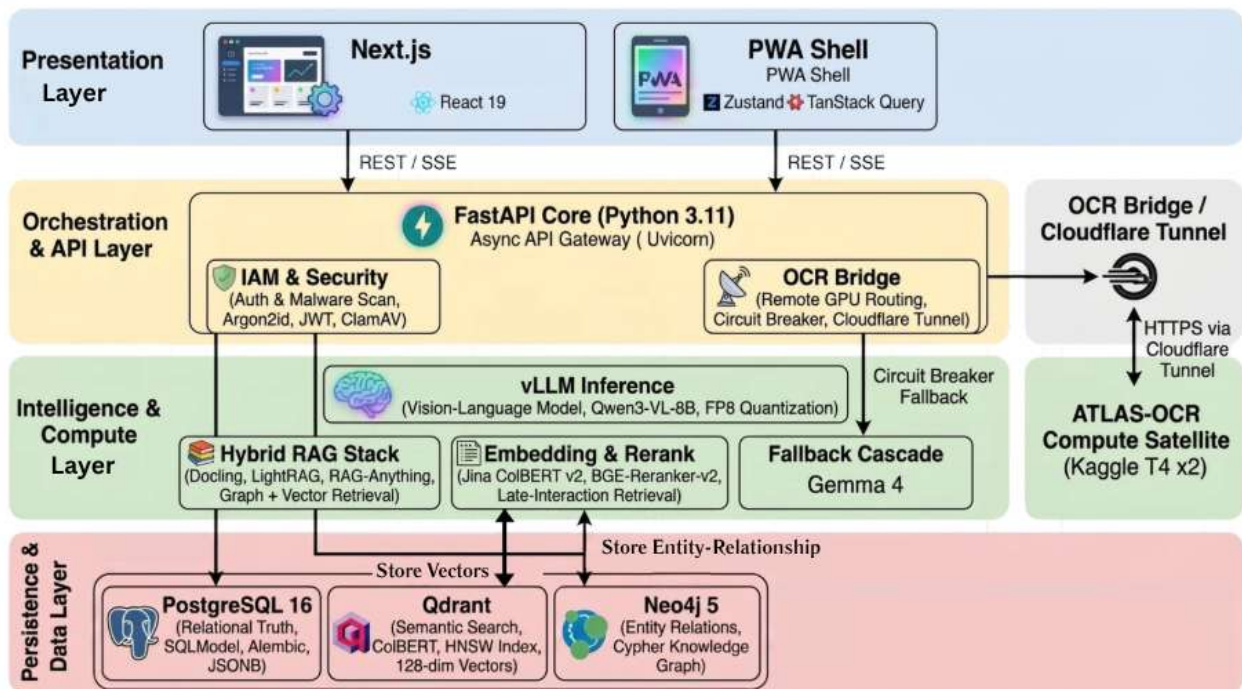
- ✓ 1. LINT
- ✓ 2. TYPE-CHECK
- ✓ 3. UNIT TESTS
- ✓ 4. PEER REVIEW
- ✓ 5. DEMO
- ✓ 6. DOCUMENTATION



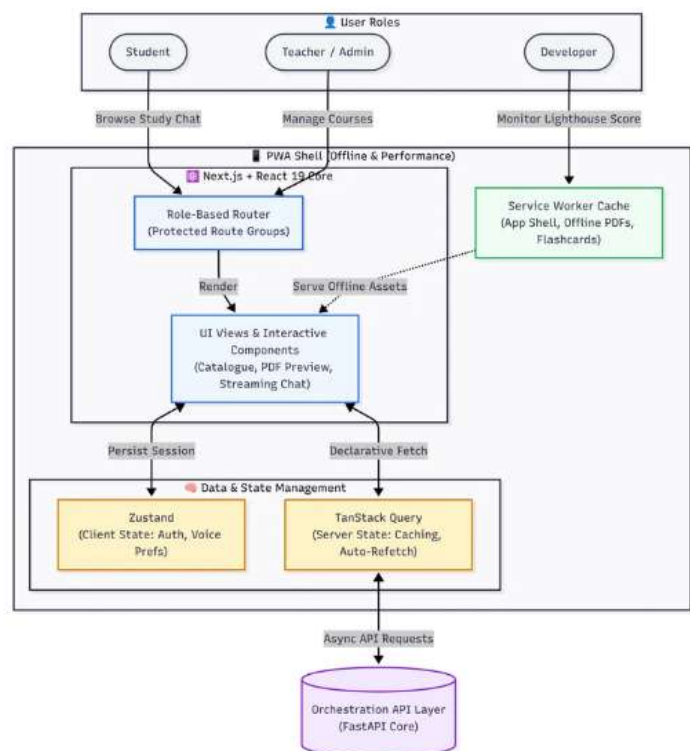


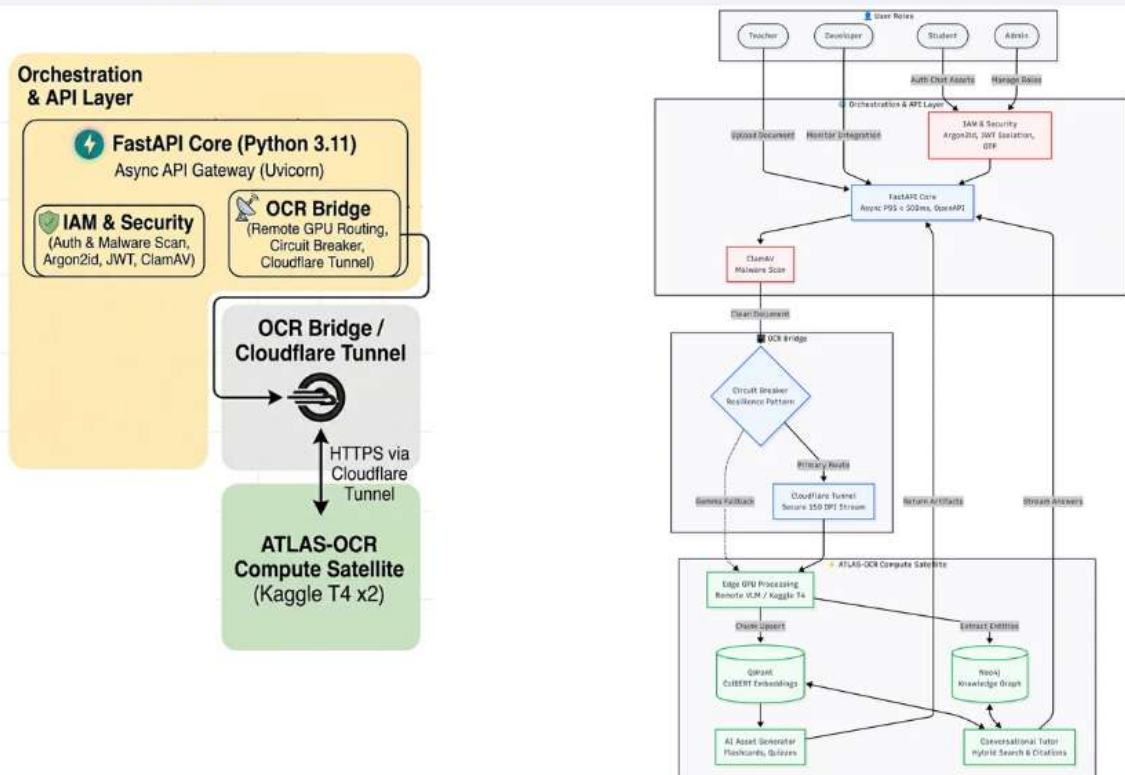


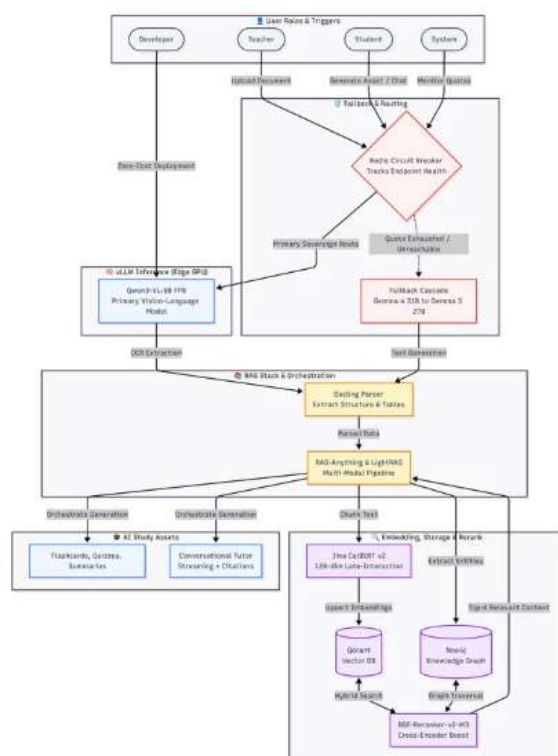
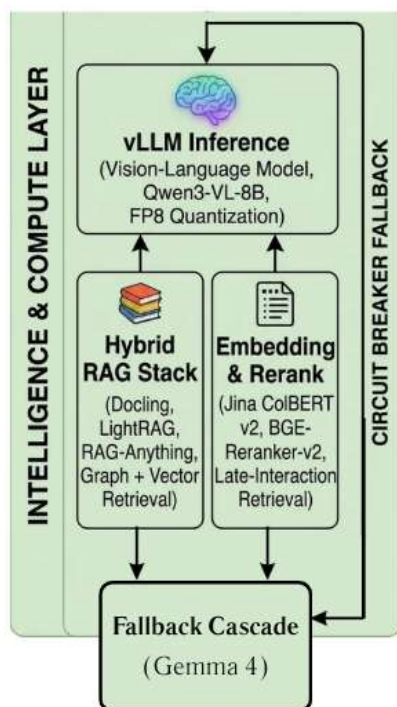
System Architecture

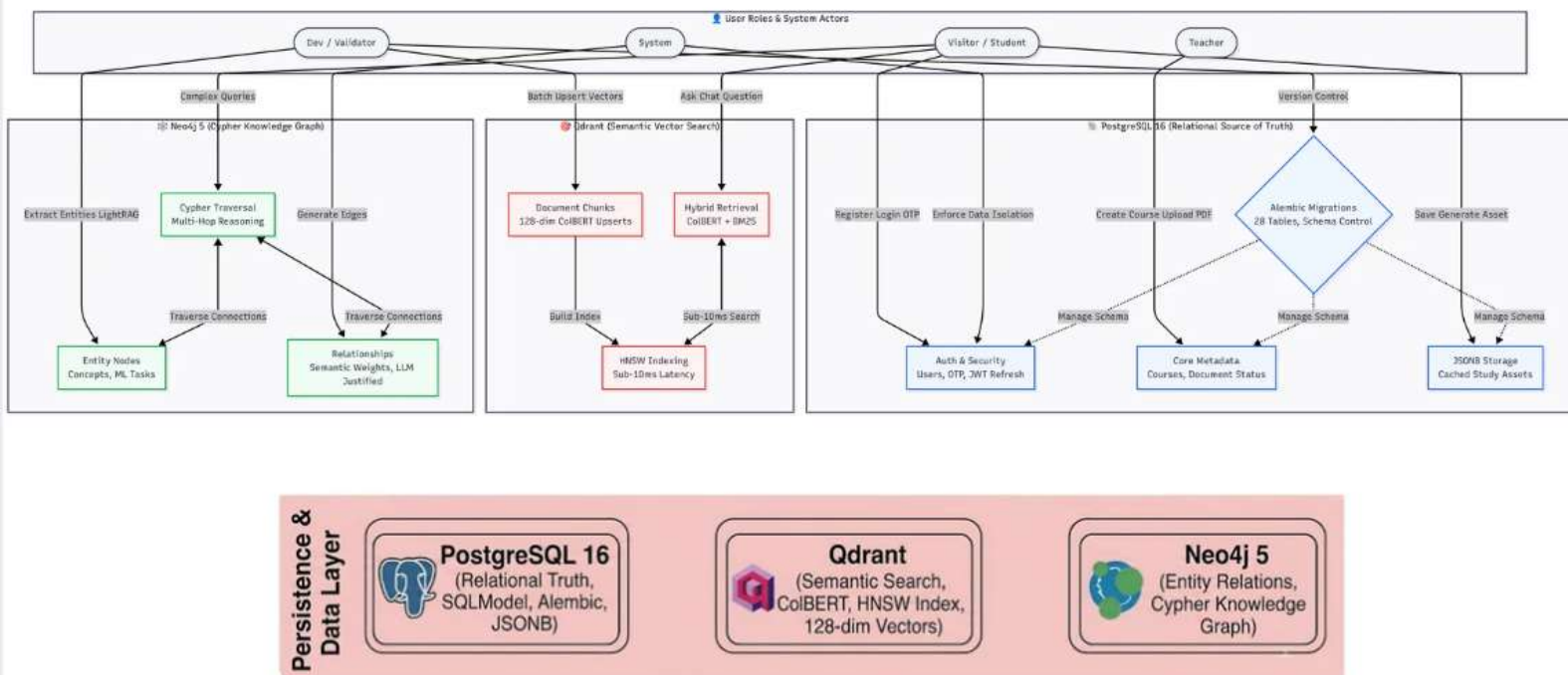


Presentation Layer

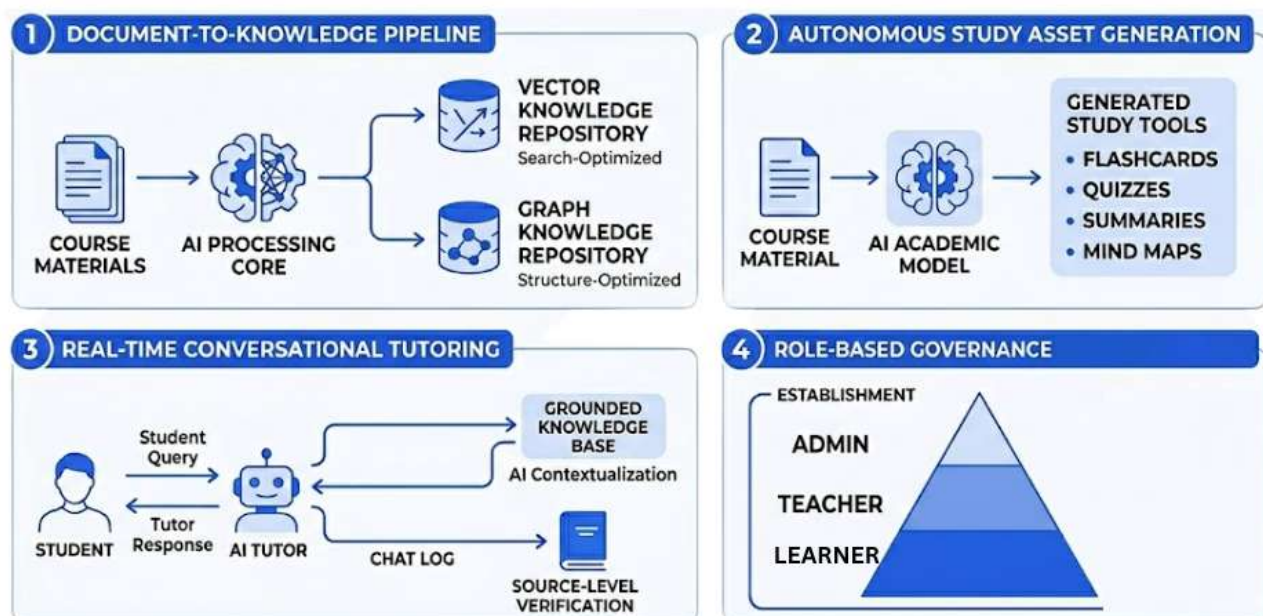








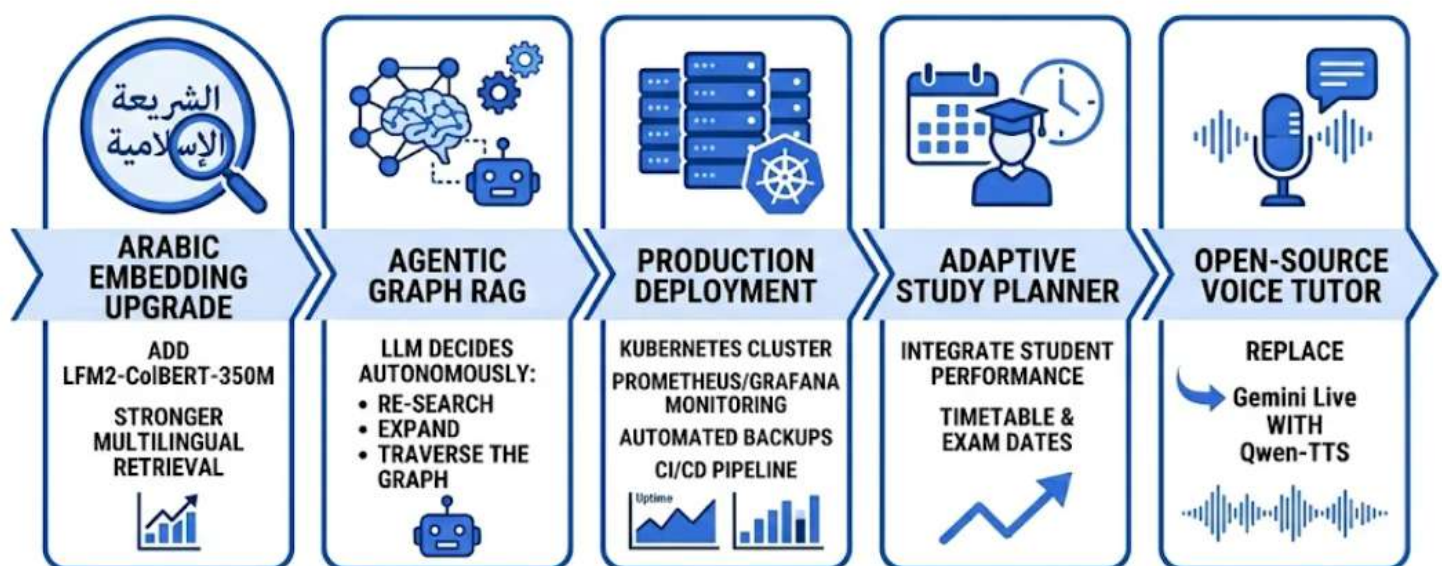
Project Deliverables



Verification Campaign

LEVEL	TOOL	COUNT	FOCUS
1. UNIT (backend)	Pytest	48 TESTS	Models, auth logic , validation
2. INTEGRATION	Pytest (API client)	15 TESTS	API endpoint contracts, Database CRUD
3. END-TO-END	Playwright	5 TESTS	Critical user journeys (login →upload → generate asset → search ...)
4. PERFORMANCE	Manual load script	3 SCENARIOS	Response time under load (GET,search, streaming)
5. FUNCIONAL VALIDATION	Manual review Gemini 3.1 Pro Extended	10 PDFS 10 QUESTIONS	OCR accuracy Fact retrieval and multi-paragraph reasoning

Roadmap & Perspectives



General Conclusion

ATLAS (Aggregated Learning & Academic System)
demonstrates that an institutional platform combining document intelligence,
AI-generated study tools, and role-based governance can be built by a small team,
at zero recurring cost, using only open-source technologies.